

SIMATS ENGINEERING

ASSESSMENT TOOL 3

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COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Introduction

Star and bus are two different network topologies used to connect multiple devices. In a star topology, every device has a separate connection to a central networking device, usually a switch or hub. In a bus topology, all devices share a single main communication cable called the backbone.

Star Topology

In a star topology, all computers are directly connected to a central switch or hub.

For example:

PC1 → Switch ← PC2

PC3 → Switch ← PC4

When PC1 wants to communicate with PC2, the data first reaches the central switch. The switch then forwards the data to the appropriate destination.

Bus Topology

In bus topology, all devices are connected to one common backbone cable.

PC1 — PC2 — PC3 — PC4

Data transmitted by one device travels along the shared cable. The intended destination receives the data, while other devices ignore it.

Comparison

Parameter	Star Topology	Bus Topology
Basic structure	Centralized	Linear/shared
Central device	Switch or hub required	Not required
Cabling	More cable required	Less cable required
Initial cost	Higher	Lower
Reliability	High	Low
Performance	High	Lower as traffic increases
Fault detection	Easy	Difficult
Scalability	Good	Limited
Maintenance	Easy	Difficult
Collision possibility	Low with switches	Higher
Network failure	Usually limited to affected device if its link fails	Backbone failure can affect the entire network

Cost Analysis

Bus topology is less expensive because all devices share a single backbone cable. There is no requirement for a central switch or hub. Star topology requires a separate cable between each device and the central switch. It also requires a switch or hub. Therefore, the initial installation cost is higher.

However, the higher initial cost of star topology can be justified because it provides better performance, easier troubleshooting, and easier expansion.

Reliability Analysis

Star topology is more reliable than bus topology for modern LANs. If the cable connecting one computer to the switch fails, only that particular computer is disconnected.

In bus topology, the backbone cable is shared by all devices. If the backbone cable is damaged, communication throughout the network can be interrupted.

Performance Analysis

Star topology generally provides better performance because a switch can intelligently forward frames to the intended device. Separate links also reduce the problems associated with sharing a single communication medium.

In bus topology, all devices share the same communication channel. As the number of devices and amount of traffic increase, network performance can decrease.

Example

A modern college computer laboratory normally uses a star topology because each computer can be connected to an Ethernet switch.

Bus topology was more common in older Ethernet networks using coaxial cable and is now rarely used for modern LAN installations.

Conclusion

Although bus topology is cheaper and simpler, star topology is generally preferred because it provides better performance, reliability, scalability, and maintenance.

2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

Star Topology

In star topology, every device connects to a central switch.

PC1 → Switch ← PC2

PC3 → Switch ← PC4

The switch acts as the central point of communication.

Comparison

Parameter	Mesh Topology	Star Topology
Fault tolerance	Very high	Moderate to high
Alternate paths	Multiple	Limited
Central dependency	No single central device	Yes
Link failure	Usually communication continues through another path	Affected device may lose connection
Switch failure	Not applicable as a central dependency	Can affect all connected devices
Reliability	Excellent	Good
Cost	Very high	Moderate
Complexity	Very high	Low
Maintenance	Difficult	Easy

Fault Tolerance in Mesh

The main advantage of mesh topology is redundancy.

Suppose device A normally communicates with device D through one link. If that link fails, another available path can be used.

This makes mesh topology suitable for environments where network availability is extremely important.

Fault Tolerance in Star

Star topology also provides some level of fault isolation.

If one computer's cable fails:

PC1 ✗ — Switch — PC2, PC3, PC4

PC1 loses connectivity, but PC2, PC3, and PC4 can continue communicating.

However, if the central switch fails:

PC1 — ✗ — PC2

PC3 — ✗ — PC4

all connected devices may lose network connectivity.

Cost Consideration

Mesh topology requires many cables and network interfaces. For **n devices**, a full mesh requires:

$$\text{Number of links} = n(n - 1) / 2$$

For 5 devices:

$$5(5 - 1) / 2 = 10 \text{ links}$$

For 10 devices:

$$10(10 - 1) / 2 = 45 \text{ links}$$

Therefore, the number of connections increases rapidly.

Star topology requires only one connection per device to the switch, making it much cheaper and easier to implement.

Example

Mesh topology is useful in critical communication environments where alternate paths are required.

Star topology is commonly used in offices, schools, colleges, and laboratories because it provides a practical balance between reliability and cost.

Conclusion

Mesh topology provides better fault tolerance and redundancy, but it has high cost and complexity. Star topology has lower fault tolerance than mesh but is much easier and cheaper to install and maintain.

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Introduction

Installation complexity refers to the difficulty involved in setting up the network, including cabling, device connections, configuration, and testing.

Bus topology is one of the simplest topologies to install, while mesh topology is one of the most complex.

Bus Topology Installation

Bus topology uses a single backbone cable.

Example:

Terminator — PC1 — PC2 — PC3 — PC4 — Terminator

Each device is connected to the backbone.

The number of cables and connections is relatively small.

Advantages of Installation

- Requires less cable.
- Does not require a central switch.
- Simple physical arrangement.
- Low installation cost.
- Suitable for small networks.

Disadvantages

- Backbone failure can affect the entire network.
- Adding many devices can reduce performance.
- Troubleshooting backbone faults can be difficult.
- Expansion is limited.

Mesh Topology Installation

In a full mesh topology, each device needs a direct connection to every other device.

For example:

A ↔ B ↔ C

A ↔ C

With additional devices, the number of connections increases rapidly.

Comparison

Factor	Bus	Mesh
Cable requirement	Low	Very high
Number of connections	Low	Very high
Installation time	Short	Long
Cost	Low	High
Configuration	Simple	Complex
Expansion	Difficult	Complex
Troubleshooting	Moderate/Difficult	Difficult
Reliability	Low	Very high

Analytical Comparison

Bus topology is suitable when the network is small and cost is the primary concern. However, mesh topology is preferred when reliability is more important than installation cost.

For example, adding one device to a bus network generally requires connecting it to the backbone. In a full mesh network, adding one new device requires connections to all existing devices.

If there are 5 existing devices and a sixth device is added, the new device requires 5 additional connections.

Therefore, mesh topology becomes increasingly complex as the number of devices increases.

Example

A small temporary network may use a simple topology, while a critical communication infrastructure may use mesh connectivity because network availability is more important than installation simplicity.

Conclusion

Bus topology is simple, inexpensive, and easy to install, while mesh topology provides high reliability and redundancy at the cost of increased installation complexity and expense.

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Introduction

Scalability refers to the ability of a network to grow by adding new devices, departments, buildings, or network segments without significantly affecting existing operations.

Hybrid topology generally provides greater scalability because it combines two or more different topologies.

Star Topology and Scalability

Star topology can be expanded by connecting additional devices to the central switch.

For example:

PC1

PC2 → Switch ← PC3

PC4

If the switch has unused ports, new computers can be connected easily.

However, the number of devices is limited by the switch's port capacity.

If all ports are occupied, an additional switch may be required.

Hybrid Topology and Scalability

Hybrid topology combines multiple topologies.

For example, an organization may use:

- Star topology for the HR department.
- Star topology for the Finance department.
- Star topology for the Sales department.
- Another topology for connecting different departments.

A simplified structure could be:

HR PCs → HR Switch

↓

Finance PCs → Central Network ← Sales PCs

This allows each department to be managed independently.

Comparison

Parameter	Hybrid Topology	Star Topology
Scalability	Very high	High
Flexibility	Very high	Moderate
Expansion	Flexible	Depends on switch capacity
Network structure	Multiple topology types	Centralized
Large organization support	Excellent	Good
Management	More complex	Easier
Cost	Can be high	Moderate

Advantages of Hybrid Scalability

Hybrid topology allows organizations to design different parts of the network according to their specific requirements.

For example:

- A department requiring high reliability can use mesh.
- A normal office department can use star.
- A larger backbone can use another suitable structure.

This allows the network to grow without completely replacing the existing infrastructure.

Example

Consider a university with:

- Administration department
- Computer Science department
- Electronics department
- Library
- Hostels

Each department can have its own local star network. These individual networks can then be interconnected to create a larger hybrid network.

Conclusion

Star topology provides good scalability for small and medium-sized networks. Hybrid topology provides **greater flexibility and scalability**, particularly for large organizations with multiple departments, buildings, or locations.

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Introduction

Network maintenance includes monitoring devices, identifying faults, replacing damaged cables, troubleshooting connectivity problems, and ensuring that the network operates efficiently.

The topology selected has a major effect on maintenance requirements.

Star Topology Maintenance

Star topology is relatively easy to maintain.

Each device has a separate connection to the central switch. Therefore, if one cable fails, the administrator can identify the affected connection easily.

Advantages

- Easy fault identification.
- Easy cable replacement.
- Individual device failures are isolated.
- Centralized management.
- Simple troubleshooting.

Disadvantage

The central switch is an important component. If it fails, many or all connected devices can lose connectivity.

Mesh Topology Maintenance

Mesh topology provides excellent reliability but is difficult to maintain because of its large number of connections.

A technician may need to check several links to determine the cause of a problem.

Advantages

- High redundancy.
- Multiple paths available.
- Link failure can often be bypassed.
- Excellent reliability.

Disadvantages

- Large number of cables.
- Complex configuration.
- Troubleshooting requires more time.
- Higher maintenance cost.
- Requires skilled network administrators.

Bus Topology Maintenance

Bus topology is simple in design but can be difficult to troubleshoot when the backbone has a problem.

A failure in one section of the backbone may affect multiple devices.

The administrator may have to inspect different portions of the cable to locate the fault.

Hybrid Topology Maintenance

Hybrid topology has moderate to high maintenance complexity because it combines different network structures.

For example, an organization may have a star topology in one department and another topology in a different section.

Network administrators must understand and monitor each section independently.

Overall Comparison

Parameter	Star	Mesh	Bus	Hybrid
Fault identification	Easy	Difficult	Difficult	Moderate
Troubleshooting	Easy	Complex	Moderate/Difficult	Moderate/Complex
Maintenance cost	Low/Medium	High	Low/Medium	Medium/High
Reliability	High	Very High	Low	High
Required skill	Moderate	High	Low/Moderate	High
Cable management	Easy	Difficult	Simple	Moderate/Complex

Example

In a college computer laboratory using star topology, if one student's network cable is damaged, the administrator can replace that particular cable without affecting the other computers.

In contrast, in a mesh network, the administrator may need to inspect several interconnected links when diagnosing a communication problem.

Conclusion

Star topology is generally the easiest to maintain. Mesh provides the highest reliability but requires more maintenance effort. Bus topology is simple but backbone failures can be difficult to locate. Hybrid topology provides flexibility but requires more skilled administration because different topology structures may be present.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and	Generally organized	Somewhat disorganized	Poorly structured

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
		logically structured			
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion